

Using /setblock or /fill to re-place a block entity keeps old NBT (if no NBT specified) but clears inventories, even when the command fails

MC-117574 Commands

Status	Resolved / Fixed
Affected	Minecraft 1.12 Pre-Release 3, Minecraft 1.12 Pre-Release 4, Minecraft 1.12 Pre-Release 5, Minecraft 1.12 Pre-Release 6, Minecraft 1.12 Pre-Release 7, Minecraft 1.12 (+178 more)
Fixed in	25w02a
Reported	2017-05-17

STEPS TO REPRODUCE

1. Launch Minecraft.
2. Create a new Creative world with default settings.
3. Run ``/setblock ~1 ~ ~ dropper[facing=up]{CustomName:"first",Items:[{id:stone,Count:1b,Slot:0b}]}``
4. Run ``/setblock ~1 ~ ~ dropper[facing=up]{CustomName:"updated",Items:[{id:stone,Count:1b,Slot:0b}]}``
5. Observe that the command fails, the dropper name remains "first," and its inventory is cleared.
6. Run ``/setblock ~1 ~ ~ command_block[facing=up]{Command:"first"}``
7. Run ``/setblock ~1 ~ ~ command_block[facing=down]``
8. Observe that the command block still contains the command "first" instead of being empty.

OBSERVED BEHAVIOUR

Actual behavior

- When ``/setblock`` or ``/fill`` targets an existing block entity without specifying block-entity NBT, the existing block entity data is retained instead of being cleared or replaced with the new block's default data.
- If the command specifies the same block state as the block already present, the command fails based on the block state comparison. However, the operation still clears the block entity's inventory, leaving other NBT unchanged.
- In Example 1, the second command fails because both droppers have ``facing=up``. The ``CustomName`` remains ``"first"`` rather than changing to ``"updated"``, while the dropper's inventory is cleared. Omitting the NBT has the same result.
- In Example 2, changing the command block's state from ``facing=up`` to ``facing=down`` does not clear its existing block entity data. The command block continues to contain ``Command:"first"`` instead of becoming empty.
- Supplying block-entity NBT in the replacement command causes the block to be placed with the specified data as expected.

EXPECTED BEHAVIOUR

Using ``/setblock`` or ``/fill`` to replace a block entity should succeed whenever the specified block state or NBT differs from the existing block, including when only NBT changes. A successful replacement should apply the provided NBT, while omitting NBT should create the block with its default data rather than retaining old data. If the command fails, the block entity and its inventory should remain unchanged.

ENVIRONMENT

Minecraft Java Edition Versions: 1.12 Pre-Release 3, 1.12 Pre-Release 4, 1.12 Pre-Release 5, 1.12 Pre-Release 6, 1.12 Pre-Release 7, 1.12, 1.12.1, 1.12.2, 17w46a, 17w47b, 17w48a, 17w50a, 18w01a, 18w03b, 18w08b, 18w11a, 18w19b, 1.13-pre1, 1.13-pre2, 1.13-pre3, 1.13-pre4, 1.13-pre5, 1.13-pre6, 1.13-pre7, 1.13-pre8, 1.13-pre9, 1.13-pre10, 1.13, 18w30a, 18w30b, 18w31a, 18w32a, 18w33a, 1.13.1-pre1, 1.13.1-pre2, 1.13.1, 1.13.2-pre1, 1.13.2-pre2, 1.13.2, 18w43a, 18w43b, 18w43c, 18w44a, 18w45a, 18w46a, 18w47a, 18w47b, 18w48a, 18w48b, 18w49a, 18w50a, 19w02a, 19w03a, 19w03b, 19w03c, 19w04b, 19w05a, 19w06a, 19w07a, 19w08a, 19w08b, 19w09a, 19w11a, 19w11b, 19w12a, 19w12b, 19w13a, 19w13b, 19w14a, 19w14b, 1.14 Pre-Release 1, 1.14 Pre-Release 2, 1.14 Pre-Release 3, 1.14 Pre-Release 4, 1.14 Pre-Release 5, 1.14, 1.14.1 Pre-Release 1, 1.14.1 Pre-Release 2, 1.14.1, 1.14.2 Pre-Release 1, 1.14.2 Pre-Release 2, 1.14.2 Pre-Release 3, 1.14.2 Pre-Release 4, 1.14.2, 1.14.3 Pre-Release 2, 1.14.3 Pre-Release 3, 1.14.3 Pre-Release 4, 1.14.3, 1.14.4 Pre-Release 1, 1.14.4

Pre-Release 2, 1.14.4 Pre-Release 3, 1.14.4 Pre-Release 4, 1.14.4 Pre-Release 5, 1.14.4 Pre-Release 6, 1.14.4 Pre-Release 7, 1.14.4, 19w34a, 19w35a, 19w36a, 19w37a, 19w38b, 19w39a, 19w40a, 19w41a, 19w42a, 19w45b, 19w46b, 1.15 Pre-release 1, 1.15 Pre-Release 2, 1.15 Pre-release 3, 1.15 Pre-release 4, 1.15 Pre-release 5, 1.15 Pre-release 6, 1.15 Pre-release 7, 1.15, 1.15.1, 1.15.1 Pre-release 1, 1.15.2 Pre-Release 1, 1.15.2 Pre-release 2, 1.15.2, 20w06a, 20w07a, 20w08a, 20w09a, 20w10a, 20w11a, 20w13a, 20w13b, 20w15a, 20w17a, 20w18a, 20w19a, 20w20a, 20w20b, 20w21a, 20w22a, 1.16 Pre-release 1, 1.16 Pre-release 2, 1.16 Pre-release 3, 1.16 Pre-release 4, 1.16 Pre-release 5, 1.16 Pre-release 6, 1.16 Pre-release 7, 1.16 Pre-release 8, 1.16 Release Candidate 1, 1.16, 1.16.1, 20w27a, 20w28a, 20w29a, 20w30a, 1.16.2 Pre-release 1, 1.16.2 Pre-release 2, 1.16.2 Pre-release 3, 1.16.2 Release Candidate 1, 1.16.2 Release Candidate 2, 1.16.2, 1.16.3 Release Candidate 1, 1.16.3, 1.16.4 Pre-release 1, 1.16.4 Pre-release 2, 1.16.4 Release Candidate 1, 1.16.4, 20w46a, 20w48a, 20w51a, 21w03a, 1.16.5, 21w05a, 21w05b, 21w06a, 21w07a, 21w11a, 1.17 Pre-release 2, 1.17, 23w07a, 1.19.4, 1.20 Pre-release 6, 1.20.2, 1.20.4, 24w13a, 24w44a, 1.21.3, 1.21.4 (includes snapshots)

ORIGINAL DESCRIPTION

The bug

When using `/setblock` or `/fill` to replace a block entity, this keeps the old NBT (if no NBT is specified), but clears the inventories, even if the command fails.

Additionally, the command will fail if the specified block state matches the existing one, even if there is new NBT to apply. The result is only inventories clearing.

How to reproduce

- Example 1:

```
/setblock ~1 ~ ~ dropper[facing=up]{CustomName:"\"first\""},Items:[{id:stone,Count:1b,Slot:0b}]}
```

This places a dropper as expected. Now run the following command:

```
/setblock ~1 ~ ~ dropper[facing=up]{CustomName:"\"updated\""},Items:[{id:stone,Count:1b,Slot:0b}]}
```

Expected behavior: command succeeds (because NBT differs), dropper updates name.

Actual behavior: command fails (because the blockstates are the same), dropper does not update name, inventory clear.

Note: Leaving off the NBT in the last command has the same effect.

- Example 2:

```
/setblock ~1 ~ ~ command_block[facing=up]{Command:"first"}
```

This places a command block as expected. Now run the following command:

```
/setblock ~1 ~ ~ command_block[facing=down]
```

Expected behavior: command block is empty

Actual behavior: command block contains the command "first".

Note: including NBT in the last command successfully places a new block as expected.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-117574>